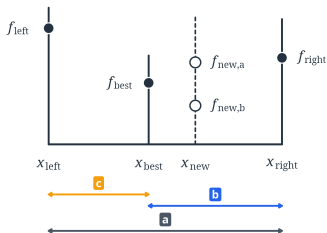
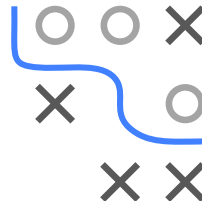


# Optimization in Machine Learning

## Univariate optimization

## Golden section search



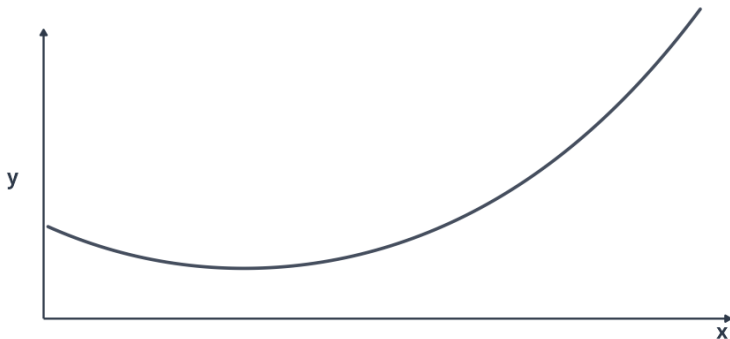
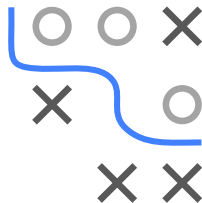
### Learning goals

- Simple nesting procedure
- Golden section search

Slides based on [Press et al. 1992](#) (Ch. 10.1)

# UNIVARIATE OPTIMIZATION

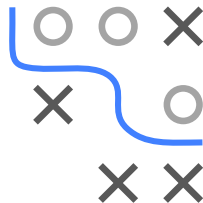
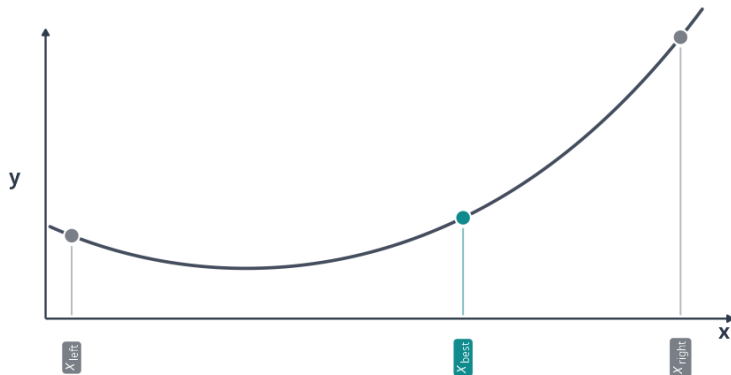
- Setting: optimize univariate  $f : \mathbb{R} \rightarrow \mathbb{R}$  over interval  $(x_{\text{left}}, x_{\text{right}})$
- Assumption:  $-f$  is unimodal and interval  $(x_{\text{left}}, x_{\text{right}})$  (called a **bracket**) contains the minimum
- Goal: iteratively shrink the bracket (**bracketing**); gradient-free, works for black-box  $f$



# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

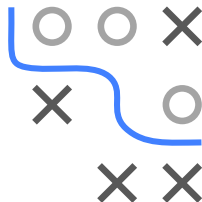
Always maintain three points: left, right, and current best



# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

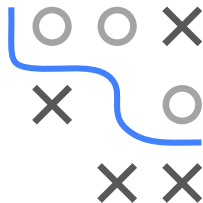
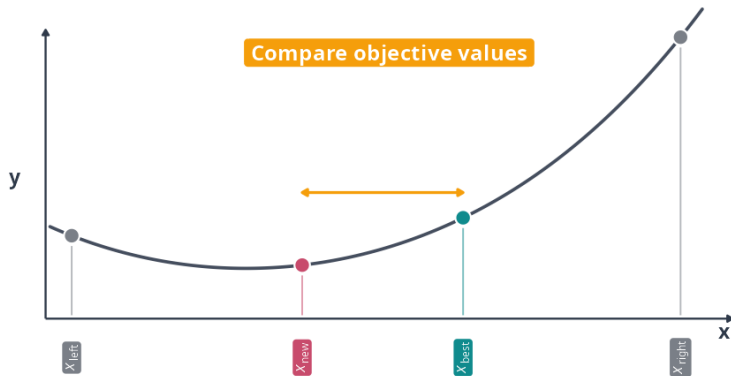
Propose random interval point (def. optimal new proposal later)



# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

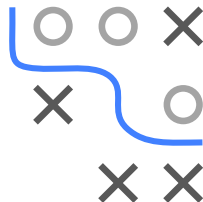
Compare proposal against current best



# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

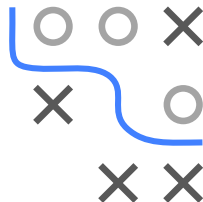
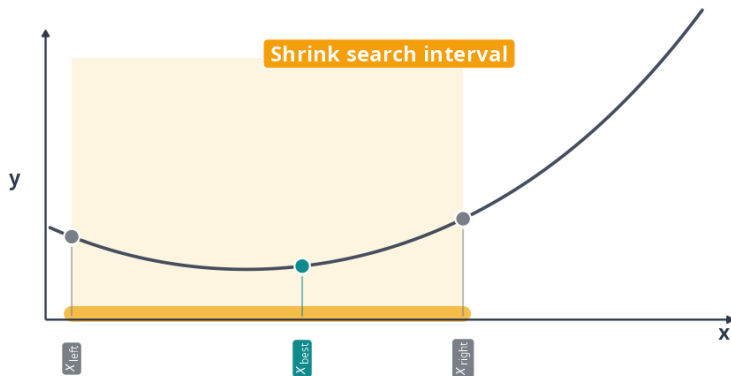
If it is better: proposal becomes current best



# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

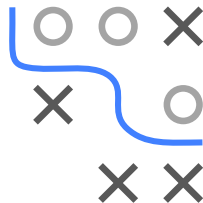
New search interval: around current best



# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

Propose a random point

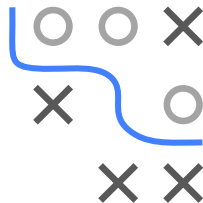
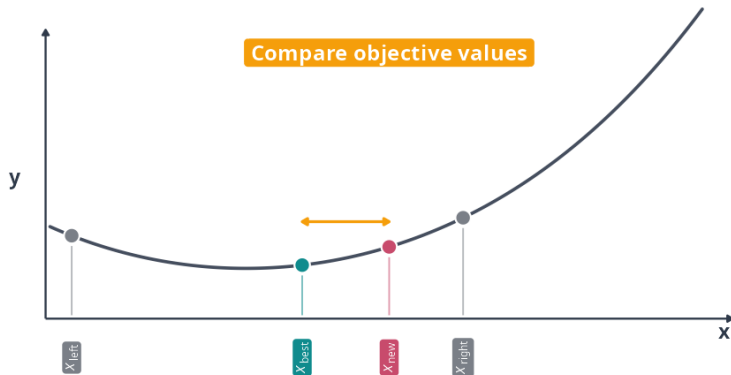




# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

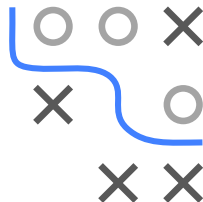
Compare proposal against current best



# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

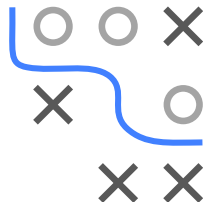
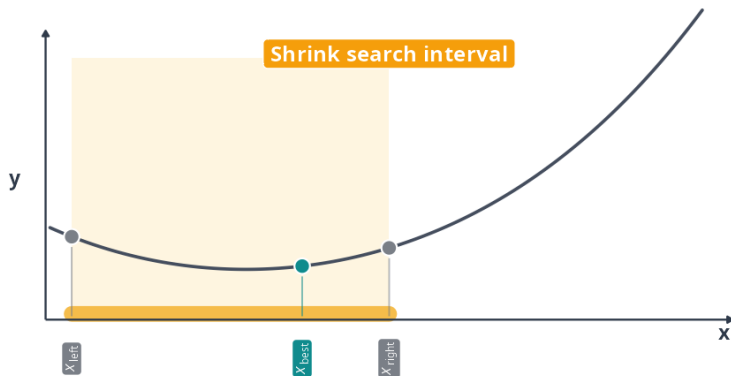
If it is better: proposal becomes current best



# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

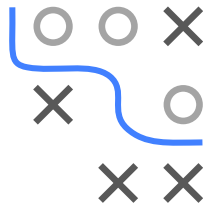
New search interval: around current best



# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

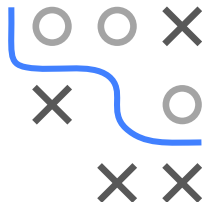
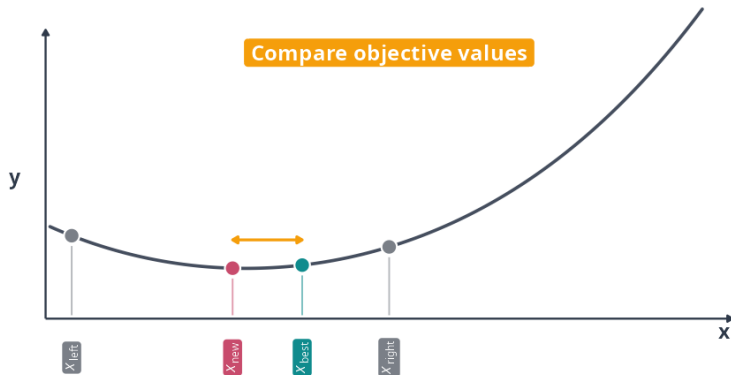
Propose a random point



# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

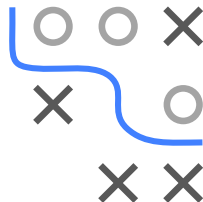
Compare proposal against current best



# SIMPLE NESTING PROCEDURE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

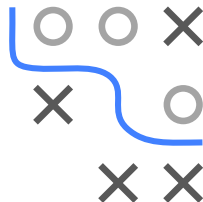
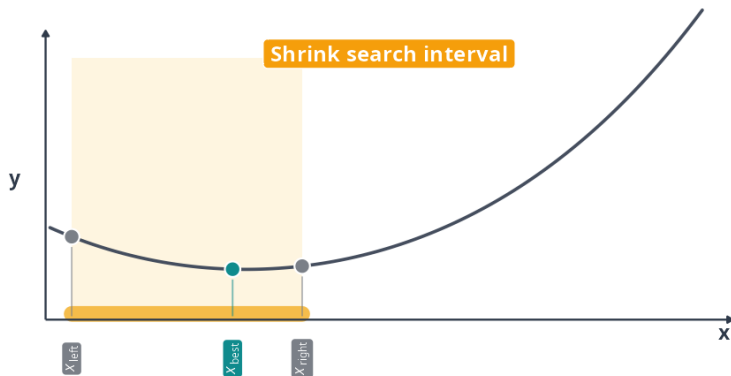
If it is better: proposal becomes current best



# SIMPLE NESTING PROCEDURE

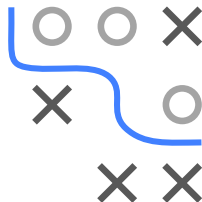
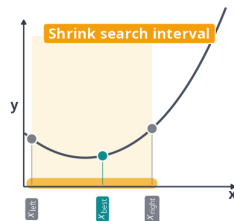
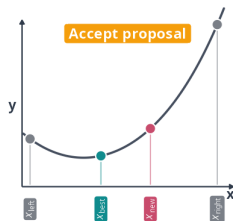
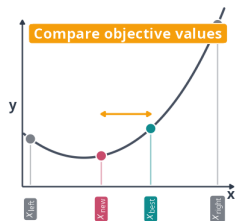
Let  $f : \mathbb{R} \rightarrow \mathbb{R}$

New search interval: around current best



# SIMPLE NESTING PROCEDURE

- Initialization: Search interval  $(x_{\text{left}}, x_{\text{right}})$ ,  $x_{\text{left}} < x_{\text{right}}$
- Choose  $x_{\text{best}}$  randomly
- For  $t = 0, 1, 2, \dots$ 
  - Choose  $x_{\text{new}}$  randomly in  $[x_{\text{left}}, x_{\text{right}}]$
  - If  $f(x_{\text{new}}) < f(x_{\text{best}})$ :
    - $x_{\text{best}} \leftarrow x_{\text{new}}$
  - New interval: points around  $x_{\text{best}}$

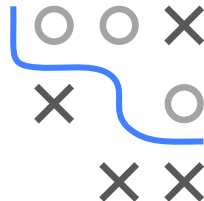
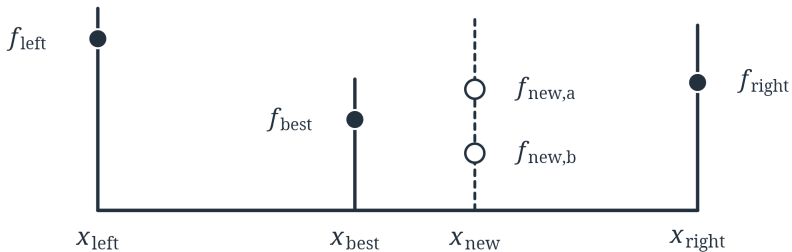




# GOLDEN SECTION SEARCH

Key question: How can  $x_{\text{new}}$  be chosen better than randomly?

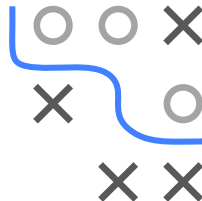
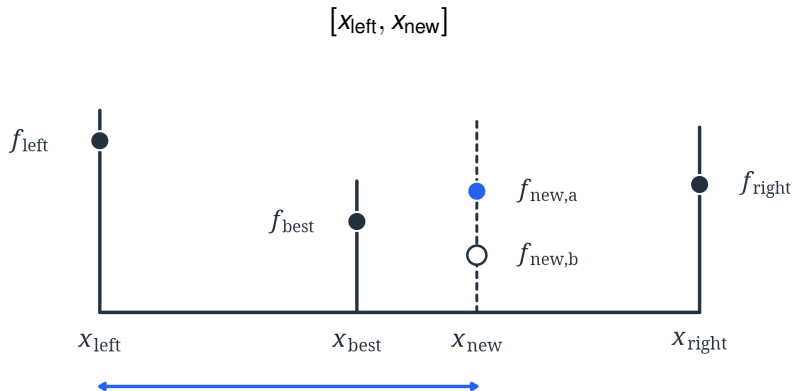
- Principle 1: Always in bigger subinterval to maximize reduction
- Principle 2:  $x_{\text{new}}$  symmetrically to  $x_{\text{best}}$  for uniform reduction



Consider two hypothetical outcomes for  $x_{\text{new}}$ :  $f_{\text{new},a}$  and  $f_{\text{new},b}$

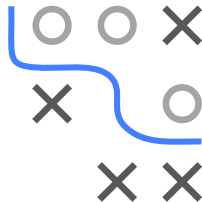
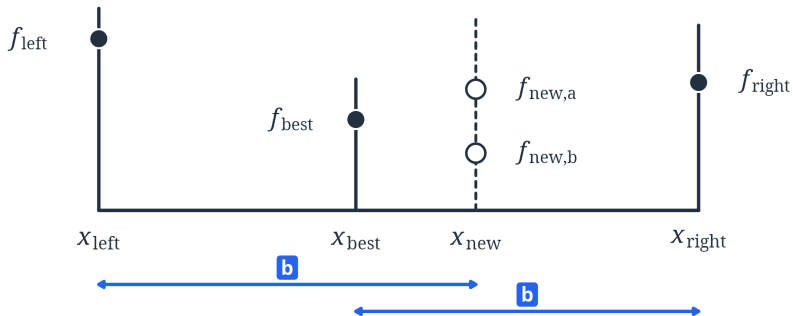
# GOLDEN SECTION SEARCH

If outcome is  $f_{new,a}$ ,  $x_{best}$  remains best and we search around  $x_{best}$ :



# GOLDEN SECTION SEARCH

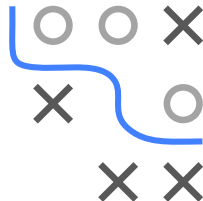
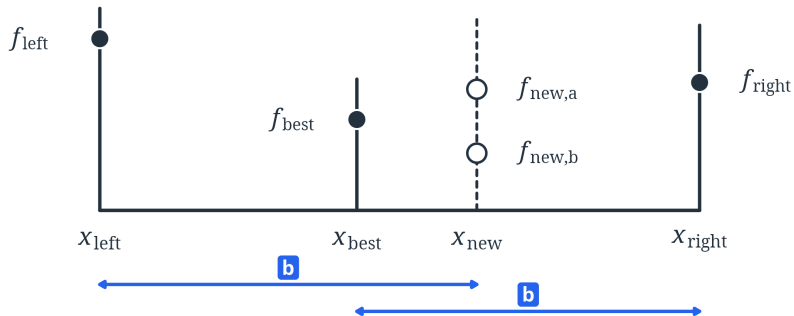
If  $f_{new,b}$  is outcome,  $x_{new}$  becomes best point and search around  $x_{new}$ :

 $[x_{\text{best}}, x_{\text{right}}]$ 

# GOLDEN SECTION SEARCH

For uniform reduction the two potential intervals must be equal-sized:

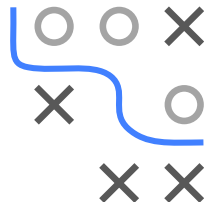
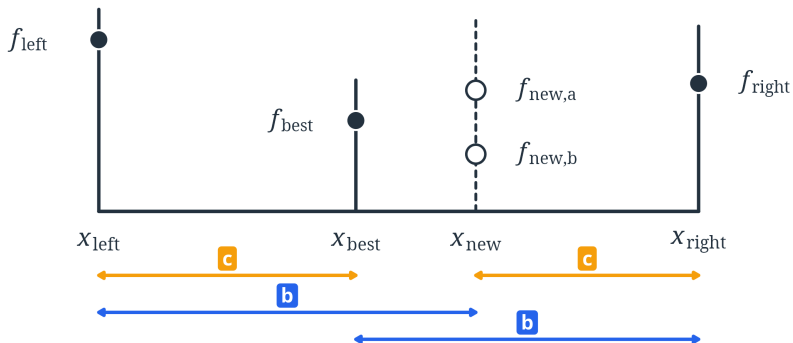
$$b := x_{\text{right}} - x_{\text{best}} = x_{\text{new}} - x_{\text{left}}$$



# GOLDEN SECTION SEARCH

One iteration ahead: require again the intervals to be of same size.

$$c := x_{\text{best}} - x_{\text{left}} = x_{\text{right}} - x_{\text{new}}$$



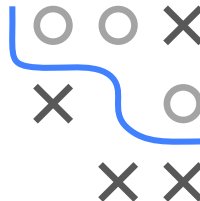
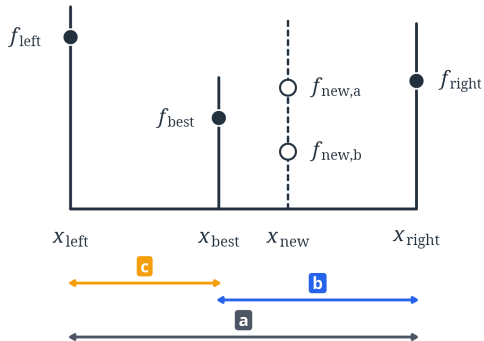
## GOLDEN SECTION SEARCH

To summarize, we require:

$$a = x_{\text{right}} - x_{\text{left}}$$

$$b = x_{\text{right}} - x_{\text{best}} = x_{\text{new}} - x_{\text{left}}$$

$$C = X_{\text{best}} - X_{\text{left}} = X_{\text{right}} - X_{\text{new}}$$



# GOLDEN SECTION SEARCH

- We require same percentage improvement in each iteration
- For  $\varphi$  reduction factor of interval sizes ( $a$  to  $b$ , and  $b$  to  $c$ )

$$\varphi := \frac{b}{a} = \frac{c}{b}$$

$$\varphi^2 = \frac{b}{a} \cdot \frac{c}{b} = \frac{c}{a}$$

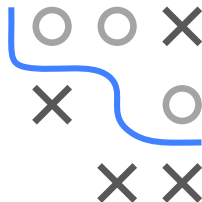
- Divide  $a = b + c$  by  $a$ :

$$\frac{a}{a} = \frac{b}{a} + \frac{c}{a}$$

$$1 = \varphi + \varphi^2$$

$$0 = \varphi^2 + \varphi - 1$$

- Unique positive solution is  $\varphi = \frac{\sqrt{5}-1}{2} \approx 0.618$



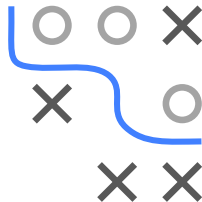
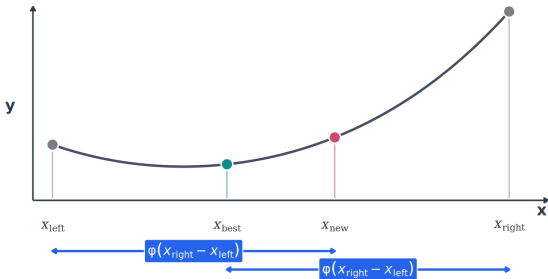
# GOLDEN SECTION SEARCH

- Place  $x_{\text{best}}$  and  $x_{\text{new}}$  at  $\varphi \approx 0.618$  from each endpoint:

$$\begin{aligned} x_{\text{best}} &= x_{\text{right}} - \varphi(x_{\text{right}} - x_{\text{left}}) \\ &= x_{\text{left}} + (1 - \varphi)(x_{\text{right}} - x_{\text{left}}) \end{aligned}$$

- By symmetry:

$$\begin{aligned} x_{\text{new}} &= x_{\text{left}} + \varphi(x_{\text{right}} - x_{\text{left}}) \\ &= x_{\text{right}} - (1 - \varphi)(x_{\text{right}} - x_{\text{left}}) \end{aligned}$$





# GOLDEN SECTION SEARCH — TERMINATION

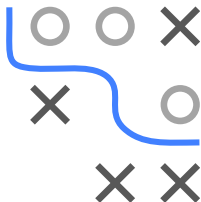
- Simple criterion: stop when bracket width falls below  $\tau$ :

$$|x_{\text{right}} - x_{\text{left}}| < \tau$$

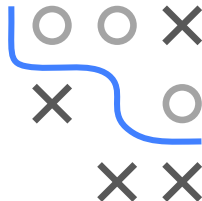
- Issue:  $\tau$  is absolute — too loose near a minimum close to 0, too tight for a minimum far from 0
- Scale-invariant alternative [Press et al. 1992](#):

$$|x_{\text{right}} - x_{\text{left}}| \leq \tau(|x_{\text{best}}| + |x_{\text{new}}|)$$

where  $|x_{\text{best}}| + |x_{\text{new}}|$  serves as a proxy for the scale of the minimum, stopping when the bracket is small relative to the minimum's location



# GOLDEN SECTION SEARCH — CONVERGENCE



- Safe method: always converges (given  $-f$  is unimodal on  $(x_{\text{left}}, x_{\text{right}})$ )
- (Slow) linear convergence: bracket shrinks by factor  $\varphi \approx 0.618$  per iteration (by construction, see derivation above)